PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION
FORM TWO MOCK EXAMINATION

Code: 032 **CHEMISRTY**

TIME: 2:30 Hours **MAY 2024**

INSTRUCTIONS:

1. This paper consists of sections A,B and C with a total of ten(10) questions
2. Answer ALL questions
3. Write your name and school name in the spaces provided
   Atomic masses H=1, C= 12, O= 16, Na= 23, Ca= 40

1. For each of the following items (i) – (x) choose the correct answer from the given alternatives and write its letter besides the item number in the box provided below;

(i) The Brownian movement is taken to be an evidence of the:-

A. Newton's theory
B. Theory of colloidal suspension.
C. Kinetic theory of behaviour of substance.
D. Theory of association of water molecules.
E. Theory of ionization.

(ii) When oxides of most non-metal are in water

A. Dilute neutral solutions are formed.
B. Alkaline solutions are formed.
C. Acidic solutions are formed.
D. Concentrated hydroxides are forms.
E. Amphoteric hydroxides are formed.

(iii)The bunsen burner flame produces much soot which is the correct conclusion:-

A. The air hole is closed.
B. The air hole is open.
C. The burner gas jet is big
D. The gas supply is poor.
E. The burner gas jet is small.

(iv)Oxygen gas can be produced in a larger scale by

A. Condensation of air of liquefied air.
B. Condensation of air.
C. Liquefaction of steam.
D. Fractional distillation of liquefied air.
E. Evaporation of liquefied air.

(v) Water is referred to as the universal solvent what does this mean?

A. Water dissolves both organic and in organic solutes.
B. Water exists in three states of matter than any other liquid.
C. Water is used more domestically than any other liquid.
D. Water dissolves more substances than any other liquids.
E. Water is neither acids nor basic compared other liquids.

(vi) Which of the following cannot be used to prepare hydrogen gas?

A. Reaction of water with carbon at room temperature.
B. Reaction of dilute acids with zinc.
C. Electrolysis of water.
D. Reaction of water with certain metals.

(vii) The main aim of destructive distillation of coal is:-

A. Removal of oxygen in the atmosphere
B. Removal of volatile matter.
C. Addition of volatile matter.
D. Addition of oxygen in the furnace.

(viii) A good fuel is determined by.....

A. Energy value.
B. High content of non –combustible material.
C. Scarcity of the fuel.
D. High production of carbon dioxide.

(ix) Which of the following statements about water is not true?

A. Water is used in industry as a coolant.
B. Water is used as a raw material to produce hydrogen.
C. Water is used as a solvent.
D. Water is used as a raw material to produce carbon.

(x) Identify the process used to remove contaminant from water.

A. Water purification.
B. Sedimentation.
C. Electrolysis.
D. Contamination.

i	ii	iii	iv	**v**	vi	vii	viii	ix	**x**

2. Match the items in list A with the responses in list B by writing the letter of the correct responses beside the item number in the box provided below.

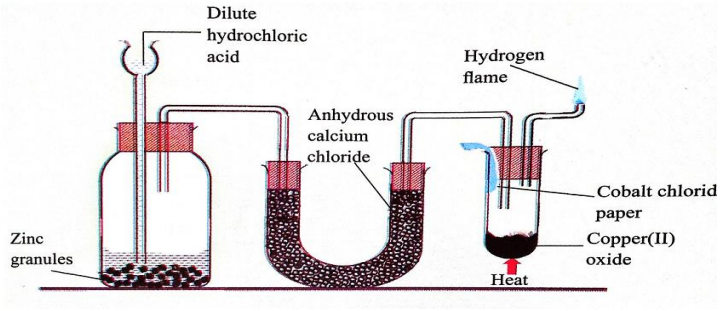
LIST A	**LIST B**
(i) Class C fires.	A. Use wet chemical extinguisher
(ii) Combustible.	B. Fire resulted from fat and oil.
(iii) Class F fires	C. Materials that produce heat on burning.
(iv) Fire triangles	D. Fire cause by flammable gases.
(v) Class B fires	E. Used to refer items needed to start fire.
	F. Fire caused by flammable liquid.
	G. Fire caused by flammable solids.
	H. Fire caused by electrical appliances

LIST A	i	ii	iii	iv	v
LIST B					

SECTION B (70 Marks)

Answer **all** questions.

3. (i)Study the diagram below and answer the questions that follow;



(a) What is the colour of the solid product? .....

(b) Name the products formed……..

(c) What is the role of the following chemicals?

(i) Hydrochloric acid and Zinc granules…….

(ii) Anhydrous calcium chloride……

(iii)Cobalt chloride paper……...

(ii) When green grass was crushed with toluene. (Methyl benzene) a green solution was formed – what method of separation can be used to show that this solution containing at four different substances? .....

4. (a) Explain the stages of urban water treatment……

……

……

……

(b) Explain why air is said to be a mixture of gases……

……

……

(c) A student wanted to prepare oxygen, by heating an oxide of mercury. His teacher did not recommend the method, what do you think could be the reason?

.....

.....

5.(a) Alex wanted to test for substances he could use to test for the presence of water in a certain compound. She put some hydrated copper (11) sulphate on a watch glass and added a few drops of distilled water to it (there was no charge in colour)

(i) Why there was no change in the colour of the copper (II) sulphate?

……

(ii) Name two substances he could have used in place of copper (II) sulphates to observe the required colour change. ……..

(b) Write the chemical symbol of the following elements

(i) Sodium……
(ii) Magnesium…….
(iii)Chlorine……..

6.(i) Give reasons to support each of the following statements;

(a) A laboratory should be equipped with working fire extinguishers
…….

(b) A laboratory should have large windows.
…….

(c) Chemicals which are not correctly labelled should not be used in the laboratory.
……

(ii)Explain the importances of changing one states of matter to another. Three (3) points

i. ……..
ii. ……..
iii. ……

(iii) With the aid of diagrams, illustrate how a portable fire extinguisher should be used.

7. (a) Explain why petroleum and coal are non-renewable sources of energy
……

(b)Oxygen gas can be prepared by thermal decomposition of potassium chloride using manganese dioxide as a catalyst.

(i) Draw well diagram to show how oxygen is prepared using this method.

(ii) Write word and chemical equation for this reaction.

…….

(c)Explain the importance of chemical symbols.

…….

8. (a) What are the warning symbol you would expect to see on

(i) A can of petrol…….

(ii) A bottle of concentrated sulphuric acid……

(b) Nekumo is a form one student at open school accidentally he mixed sulphur and iron fillings.

(i) Suggest an appropriate method of separation you would advise him to use to separate the mixture…….

(ii) Give a reason for the choice of your answer……..

(c) (i) Mention the necessary correlation for rusting to occur.

……
……..

(iii)Mention two (2) effects of rusting

……
…….

9.(a) How to provide first aid to a person affected with electric shock.

……
……
……
……
……

(b) Differentiate mixtures and compounds. Four (4) points.

i. …….
ii. ……
iii. ……
iv. ……..

SECTION C

10. (a) Draw a well labeled diagram of a small laboratory heat source consisting of vertical metal tube connected to a gas source.

(b) Describe how you will light a device drawn above

(i) .....
.....

(ii) .....
.....

(iii).....
.....

(iv).....
.....

(c) Give four differences between a flame produce when the air holes of a bunsen burner are open and the one produced when the air holes are closed.

Flame when air holes open	Flame when air holes closed
(i)	
(ii)	
(iii)	
(iv)	